Engineering Applications of Artificial Intelligence 162 (2025) 112620



Contents lists available at ScienceDirect

Engineering Applications of Artificial Intelligence

journal homepage: www.elsevier.com/locate/engappai





Causally explainable artificial intelligence on deep learning model for energy demand prediction

Gatum Erlangga, Sung-Bae Cho *

Dept. of Computer Science, Yonsei University, 50 Yonsei-ro, Sudaemoon-gu, Seoul, 03722, South Korea

ARTICLE INFO

Keywords:
Explainable artificial intelligence
Causal explanation
Convolutional neural network-long short-term memory
Neural network
Bayesian network
Power demand prediction

ABSTRACT

Accurate power demand prediction is essential for energy management in the energy sector, but it is difficult due to the factors such as greenhouse gas emissions and climate change. CNN-LSTM (Convolutional neural network-long short-term memory) neural network has demonstrated impressive performance but faces limitations in explaining its prediction results. Although XAI (explainable artificial intelligence) techniques enhance understanding through feature importance, they primarily focus on correlation rather than causality among variables in the deep learning models. To address this issue, we propose a causal XAI method for CNN-LSTM neural network with attention mechanism to predict power demand. Bayesian network is employed to provide the causal explanation with domain knowledge and relationships among observed variables and deep learning parameters (e.g., class activation maps and attention weights). Experiments on two real datasets such as UCI (University of California, Irvine) individual household electricity dataset and REFIT (Regulatory Fitness and Performance programme of the European Commission) dataset show improvements of average 34.84 % and 13.63 %, respectively. It also confirms that the proposed method not only significantly outperforms state-of-the-art models in terms of prediction accuracy but also provides a causal explanation of the prediction outcome in terms of peak power usage, savings, and stability based on the observation windows, which provides actionable insight for end users to achieve power efficiency.

1. Introduction

Precise prediction of power demand is essential for efficient energy management in the power sector (Zhuang et al., 2023; Pinto et al., 2021). This is particularly crucial because electricity, unlike water or food, cannot be easily stored and must be generated in real-time to match demand (Khan and Osińska, 2023). Additionally, the need for accurate demand forecasting has intensified due to growing concerns over greenhouse gas emission and climate change, which drive the push for sustainable energy management (Rajapaksha and Bergmeir, 2022; Wenlong et al., 2023). By aligning production with consumption needs, effective demand prediction helps prevent both energy shortages and unnecessary overproduction.

End users, particularly household occupants, play a significant role in fossil fuel consumption and carbon emissions (Stern et al., 2016). One effective strategy to encourage energy savings is by increasing their awareness. When individuals become more conscious of their electricity usage, they are more likely to adopt energy-efficient behaviors (Song and Leng, 2020), such as adjusting indoor temperatures and turning off unused lights and appliances. However, these behavioral adjustments vary across different households, and their impact on overall energy efficiency depends on multiple factors. For example, houses located in the tropical area might have a different pattern than the four-season area.

Understanding the key factors that drive energy demand peaks is crucial for addressing energy challenges more effectively. Additionally, effective energy usage prediction is essential to reducing power fluctuations (Hasan et al., 2023), demonstrating the capability of AI-driven system ability to meet regulatory compliance. At large scale, the government can integrate such predictive system into a holistic approach for strategic decision-making, using the prediction result and the corresponding explanation as the consideration of future decision (Bertoldi, 2022).

Many researchers have worked on this problem by applying deep learning models to predict power demand patterns in households and receiving competitive performance metrics. Convolutional neural network-long short-term memory (CNN-LSTM) neural network is one of the powerful models to predict power consumption demand (Kim and

* Corresponding author.
E-mail address: sbcho@yonsei.ac.kr (S.-B. Cho).

https://doi.org/10.1016/j.engappai.2025.112620
Received 9 February 2025; Received in revised form 20 June 2025; Accepted 2 October 2025
Available online 10 October 2025
0952-1976/© 2025 Elsevier Ltd. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

Cho, 2019) that offers multivariate time series processing (Chung and Jang, 2022) and spatial-temporal feature extraction (Heo et al., 2021; Ijaz et al., 2022) from the observed data. Due to its complexity, however, it is hard to understand the underlying mechanism inside the model (Bu and Cho, 2023; Ribeiro et al., 2016a), which sacrifices its explainability over the accuracy (Nagahisarchoghaei et al., 2023).

Addressing this issue, we aim to enhance both explainability and interpretability in predicting household energy demand using CNN-LSTM models. According to Mavrepis, although the terms of explainability and interpretability are often used interchangeably, some researchers make a distinction between them (Mavrepis et al., 2024). Explainability in AI refers to the extent to which a model's predictions can be understood, considering the insights provided, the target users, and the necessity of these insights. It is derived from various methods including textual descriptions and the significance of features. In contrast, interpretability focuses on ensuring that the explanation aligns with the user's pre-knowledge, with the goal of making the explanation understandable and coherent. In this paper, we define the explainability as the ability to uncover the underlying causal mechanism that influences the model's predictions. This includes uncovering relevant patterns in the model's internal decision process and its corresponding input that affects the final output, providing global understanding.

We divide the prediction outcome after the observation window shown in $(t+1)$ time step into three states, which is useful to give awareness on what would happen in the future when they use certain equipment. They are: 1) peak (Alduailij et al., 2021) indicating a high use of energy consumption, which enables occupants to decide whether to use energy efficiently or keep on the high patterns depending on the external factor, for instance, season and time of use, 2) no significant change (Park et al., 2018) reflecting stable energy demand during the observation time, which explain the stable use of energy, and 3) significant decrease (Shi and Jiao, 2023) signaling a substantial reduction in energy use, enabling occupants to enhance their energy savings, especially after periods of peak usage. These categories are chosen to provide more actionable insight, reduce cognitive load, enhance clarity and usability compared to finer states, which may increase complexity and decrease interpretability.

To provide causal explanation, we employ Bayesian network (BN) as a tool. Contrary to other explanation methods such as SHapley Additive exPlanations (SHAP) (Lundberg and Lee, 2017) and Local Interpretable Model-Agnostic Explanations (LIME) (Ribeiro et al., 2016b), BN could give insight into cause-and-effect relationships and understanding why certain phenomena occur, which is important in various domains (Zhang, 2023) including power demand prediction (Buys et al., 2015; Khan, 2021). It is constructed using association rules (Bu and Cho, 2023) and structure learning for three power prediction categories. Furthermore, we analyze deep learning parameters (DLPs) to enhance interpretability. While our method is applicable to any deep learning model, in this study, we employ CNN-LSTM, one of the most widely used models for energy prediction. Based on the model's structure, we leverage class activation maps (CAMs) and attention weights (ATTs) to explain the factors influencing prediction results. For instance, a part of the activation map would be activated because of the increasing pattern in climate control devices and active power. By applying both prediction and explainability to the power consumption domain, effective consumer behavior management can be achieved (Mosannenzadeh et al., 2017).

The rest of this paper is organized as follows. Section 2 provides the related works in deep learning and explainability for power demand prediction, and Section 3 proposes a method to predict power demand with CNN-LSTM-attention network and explain the prediction results with Bayesian network with domain knowledge to improve its trustworthiness (Tang et al., 2022). Section 4 presents the experimental results, including model performance, deep learning parameters analysis, and qualitative evaluation of the explanation model. Finally, Section 5 discusses experimental results and future research directions, and Section 6 summarizes our key findings and concludes the research.

2. Related works

Understanding the key factors in model prediction is crucial (Tjoa et al., 2023) and has various applications in power demand. It includes feature selection and understanding the main point in energy usage. Table 1 summarizes the relevant studies on explainable artificial intelligence (XAI) for energy prediction, which include various prediction models and XAI methods. In machine learning, some models have a built-in ability to provide explanations based on the learned features. For example, tree-structured model like decision tree and random forest provides insight based on constructed rules to help decision making in architectural energy efficiency (Yao et al., 2024) and helps determine the key activities in energy load profiles (Satre-Meloy et al., 2020). Furthermore, a new explainable model is also being proposed. For

Table 1
Related works of model explanation for energy prediction.

Explanation Technique	Model	Objective	Dataset
Decision tree rules	Decision tree (Yao et al., 2024)	Developing a decision tree model to optimize zero energy house designs in cold regions	China solar decathlon competition in 2022
SHapley Additive exPlanations (SHAP)	Ensemble model (Cakiroglu et al., 2024)	Predicting wind power and revealing feature importance that affects the power production	Collected from General Directorate of Meteorology at station 17,112 for Çanakkale province in Turkey
Latent variables	Autoencoder (Kim and Cho, 2023)	Predicting household power demand with autoencoder architecture and explaining with latent variables	UCI household dataset
Association rule-based modular Bayesian network	CNN-LSTM with attention (Bu and Cho, 2023)	Explaining power demand prediction by modular Bayesian network constructed by association rule mining	UCI household dataset
SHapley Additive exPlanations (SHAP)	XGBoost, SVR, CART, LightGBM, LSTM (Sim et al., 2022)	Input selection using XAI method	Gas heat pump (GHP) installed in a university building in Seoul, and self-collected time and season data
SHapley Additive exPlanations (SHAP)	Artificial neural network (ANN) (Białek et al., 2022)	Predicting heat (climate control device) demand and providing validation through SHAP	Weather and heat demand record from 2015 to 2019
LIME (Local Interpretable Model-agnostic Explanations)	Elastic net regularization, Decision trees, Random forest, ANN (Amiri et al., 2021)	Explaining the effect of household travels, demographics, and neighbor data on transportation energy consumption	Opensource Household Travel Survey (HTS) dataset
Random forest feature importance	Random forest (Satre-Meloy et al., 2020)	Predicting load profiles and identifying its key activities	Self-collected load profiles and activities data

example, Kim and Cho explained power conditions using latent variables captured by an encoder-decoder architecture (Kim and Cho, 2023).

For a non-self-explanation model, an explanation part is added after the result of prediction is given, which is called *post-hoc* approach (Theissler et al., 2022). SHAP and LIME are popular *post-hoc* explanation models which can be used to explain various machine learning models. SHAP provides the ability to understand features contribution of various prediction models in individual observation (local) and global feature contributions meanwhile LIME only supports local explanation. Global feature contribution is useful to understand the significant factor of electricity demand, while local explanations provide explanations on how certain state of energy demand contribute to the current energy demand pattern (EDP) (Rajapaksha and Bergmeir, 2022).

Recent studies have significantly diversified the application of predictive models in energy demand scenarios, showing the adaptability of XAI methods across various contexts. For instance, Cakiroglu et al. developed an ensemble model aimed at predicting wind power and revealing key features that impact power production, using SHAP to highlight these important attributes (Cakiroglu et al., 2024). In addition, Bialek et al. employed an artificial neural network (ANN) to forecast the demand for climate control devices, applying SHAP to validate the model's prediction (Białek et al., 2022). Moreover, Sim et al. explored a combination of models, including XGBoost, SVR, CART, LightGBM, and LSTM, for input selection in energy consumption, leveraging SHAP to enhance the explainability of their methods (Sim et al., 2022). These examples demonstrate SHAP's ability to provide vital explanation across a variety of models and settings, highlighting its effectiveness in extracting meaningful insights from complex predictive models. However, despite its robustness in explaining various models, SHAP does not address dependency or elucidate the causal relationships between features.

However, due to complex interactions among variables in nature, Bu and Cho developed a Bayesian network based on association rule to model the causal relationships (Bu and Cho, 2023). They combined observed variables and deep learning parameters such as activation maps and attention weights to explain the current state of energy demand in a certain time window. Further analysis of the activation map and attention weights is conducted to improve the interpretation of the XAI model. Furthermore, domain knowledge is employed in the Bayesian network structure learning process to improve the trustworthiness of the structure (Tang et al., 2022).

3. The proposed method

We aim to endow the prediction model with explainability based on causal relationships among variables, for which we utilize Bayesian network to model causality. Causality could provide insight into the underlying mechanisms of power consumption patterns. This approach not only highlights the changes in certain aspects of energy use can affect another but also helps in distinguishing genuine causal relationships from spurious correlations (Richman and Roberts, 2023).

Causality, understood as the cause-and-effect relationship from parent nodes to child nodes, is modeled within Bayesian network through a directed acyclic graph (DAG). This relationship is inferred through causal discovery, which assumes relationships among variables and quantifies them using observational data (Lu et al., 2023; Nogueira et al., 2022). In this paper, we focus on three power consumption patterns: peak, no significant change, and lower than usual. With these three representative states, we can explain the basic conditions to household occupants at each time step when they activate certain power states.

3.1. Overview

The method starts with CNN-LSTM neural network to predict energy demand from time series data, as shown in Fig. 1. The architecture is organized into two components, parallel CNN and LSTM modules to extract spatial and temporal features (Section 3.2), and causal explanation module that analyzes deep learning parameters and input features using Bayesian network (Section 3.3). The prediction result is categorized into one of three states, such as 'peak warning' (Alduailij et al., 2021), 'no significant change' (Park et al., 2018), and 'lower than usual' (Shi and Jiao, 2023). Here, a statistical approach is used to detect an outlier based on the observed window in the prediction model. By looking at the state per window, energy demand patterns can be extracted based on some context, particularly observation windows that we are interested in.

The prediction triggers a Bayesian network with the causal

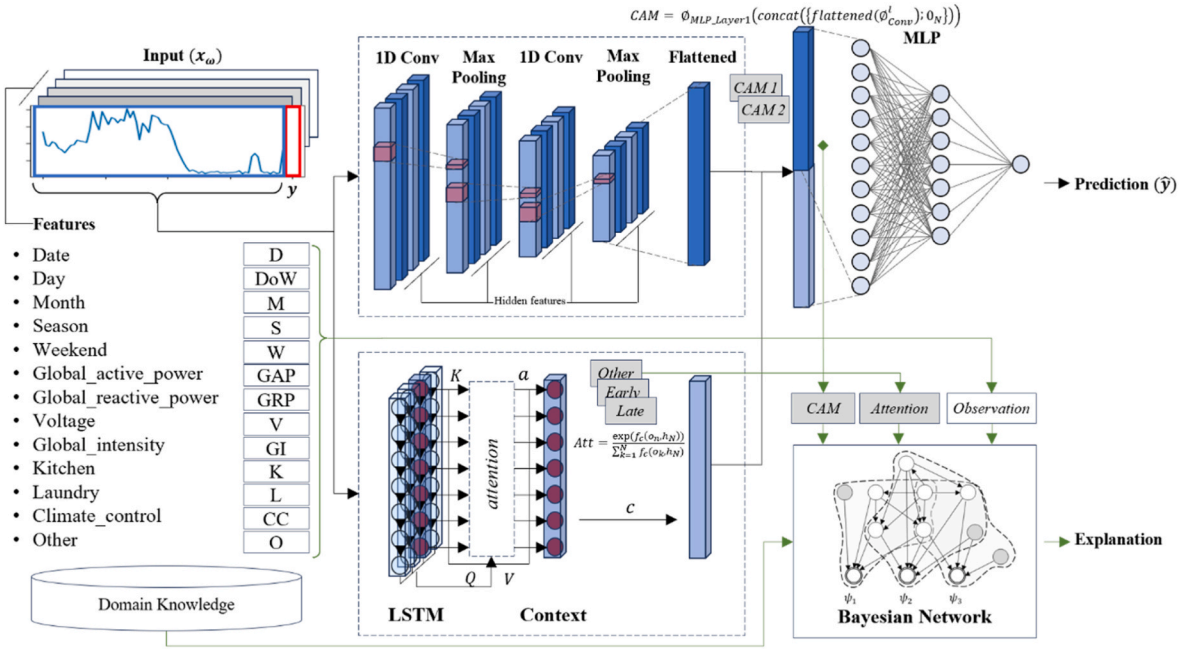


Fig. 1. Overall structure of the proposed method.

relationship among features. The BN is constructed by structure learning with association rules (Alduailij et al., 2021) and domain knowledge found in the literature. All variables in the observation data are discretized to provide not only simplicity but also interpretability and computational efficiency (Chen et al., 2017). Lastly, further analysis of deep learning parameters provides to improve the interpretability of the prediction result.

3.2. Energy prediction with CNN-LSTM with attention

We develop parallel CNN-LSTM neural network with attention to capture spatial and temporal information separately. Temporal features such as fluctuations and usage patterns are effectively captured by the CNN through recognition over short-time windows and relationship between different input features. Temporal dependencies such as long-term trends and seasonal effects are better captured by the LSTM through its ability to learn sequence history. As a result, this method minimizes the abstraction after putting LSTM in front of CNN (Zhang et al., 2021). Before feeding the data into the model, we adjust the number of input sequences into observations window ω and target label l. For instance, we set ω into 80 timestamps, and l into one step ahead. The ith-power consumption data are denoted as $x_i = \{x_{i,1}, \ldots, x_{i,n}\}$, where n is the observation window ω. After that, the pre-processed and normalized data are put into both CNN and LSTM respectively.

In convolution step, we use function $\varnothing^l_{Conv}$ to map feature x into set of feature representations learned by the convolution at layer l. This function can be stacked and produce a deep convolution layer by inserting the previous output of layer l into the function. In temporal feature extraction step, we extract two components after calculating $\varnothing^o_{LSTM}$ as the LSTM operation. For N inputs, this function produces $H = \{h_1, .., h_N\}$, $S = \{s_1, .., s_N\}$ and $O = \{o_1, .., o_N\}$, as hidden state, cell state and output of LSTM, respectively. These three components are then used to calculate attention, a proven method to help the model pay more attention to the specific data point, such as time series data. To obtain the attention value of time n, first, we calculate the attention weights through the function $f_c(o_n, h_N)$, which measures the relationship between o_n and h_N.

$$a_n = \frac{\exp\left(f_c(o_n, h_N)\right)}{\sum_{k=1}^{N} f_c(o_k, h_N)}, \text{for } t = 1, \ldots, T \tag{1}$$

$$c_n = \sum_{k=1}^{N} a_k \cdot h_k \tag{2}$$

Using equations (1) and (2), we can obtain soft attention weights and context vectors for each time of LSTM output. To combine both extracted representations from the parallel model, we simply concatenate both outputs of last layer from $\varnothing^l_{Conv}$ and learned context c_N from the LSTM attention layer, as shown in equation (3):

$$X_{combined=} concat\left(\{flattened\left(\varnothing^l_{Conv}\right); c_N\}\right) \tag{3}$$

After the spatial-temporal features have been obtained, the last step is to perform regression. Here, we use a multi-layer perceptron (MLP) to capture knowledge from the learned representation. We map this operation with function $\varnothing_{MLP}$, which transforms x into prediction label $\widehat{y}$ as shown in equation (4).

$$\widehat{y} = \varnothing_{MLP}(X_{combined}) \tag{4}$$

3.3. Causal explanation with Bayesian network

Symbol Extraction. To design the Bayesian network, we extract symbols from the observation data by discretizing the continuous value to improve the interpretability using two methods. First, we identify the prediction result out of three states, peak, no significant change, and decrease using statistical technique. S_n scale estimator is used for outlier detection, which is more robust to a noisy environment (Yaro et al., 2024), including the electricity domain. S_n scale estimator is presented by:

$$S_n = c^* median_m\{median_n\{|x_m - x_n|\}\} \tag{5}$$

where $m \in \{0, 1, .., N-1\}$, $n \in \{0, 1, ..N-1\}$, $c = 1.1926$, and $n \neq m$. Since n denotes the last observation time, n is $N-1$. Given data observation at time x_i, median at the observation window $MED(X)$, and correction parameter $\alpha = 1.4285$, the calculation of Z-score value using S_n (where $n = i$) can be obtained by equation (6). The result is then filtered using a threshold to obtain three different EDPs as shown in equation (7).

$$Z_{S_n}(x_i) = \frac{|x_i - MED(X)|}{\alpha \times S_n} \tag{6}$$

$$f_{EDP}(Z_{S_n}(x_i)) = \begin{cases} \text{"peak warning"}, Z_{S_n}(x_i) > threshold \text{ and } Z_{S_n}(x_i) > m_x \\ \text{"lower than usual"}, Z_{S_n}(x_i) > threshold \text{ and } Z_{S_n}(x_i) < m_x \\ \text{"no significant change"}, \text{otherwise} \end{cases} \tag{7}$$

Next, we discretize the data into four different states of energy use, which represent the typical use of energy from low to high (Bu and Cho, 2023). Quantile-based discretization is applied (equations (8) and (9)) to the dataset, which has already been used to quantize energy consumption data (Martinez Lagunas et al., 2021).

$$Q_r = (N+1)\frac{r}{4}, r \in 1, 2, 3 \tag{8}$$

$$f_{Usage}(x_i) = \begin{cases} \text{"Low"}, i \leq Q_1 \\ \text{"Medium"}, Q_1 < i \leq Q_2 \\ \text{"High"}, Q_2 < i \leq Q_3 \\ \text{"Very High"}, i > Q_3 \end{cases} \tag{9}$$

In addition, we perform data engineering to extract seasonal data that influence other variables, such as season, day, weekend, and month.

Since deep learning model contributes directly to the prediction result (Cunningham et al., 2023), we extract attention weights a_n from equation (1) and activation map which shows the activation of certain prediction result (Zhou et al., 2016). The activation is obtained using equation (10), where $\mathbf{0}$ denotes a zero vector with a length of N, obtaining only the last activated convolution output. Lastly, given the activation map and attention vector, K-means algorithm is used to get k groups of similar patterns (Satre-Meloy et al., 2020), selected using elbow method (Afzalan et al., 2021; Cen et al., 2022).

$$Activation\ Map = \varnothing_{MLP_Layer1}\left(concat\left(\{flattened\left(\varnothing^l_{Conv}\right); \mathbf{0}_N\}\right)\right), \mathbf{0}_N \in \mathbb{R}^N \tag{10}$$

Association Rule Mining. Next, association rule mining is conducted to extract useful relationships between symbols that are used to construct a Bayesian network. Given rule R and symbols $S = \{S_{GAP_Low}, S_{Intensity_Low}, \ldots\}$, the relationship on the given symbols is shown as an if-then rule by the left-hand side (LHS) to the right-hand side (RHS), or $LHS \rightarrow RHS$ (denoted by $R_{EDP} = \{r_1, \ldots, r_i\}$). Utilizing Apriori algorithm, it employs confidence and lift to assess the strength of the relationship beyond frequency.

$$Confidence(LHS \rightarrow RHS) = \frac{Support(LHS \cup RHS)}{Support(LHS)} \tag{11}$$

$$Lift(LHS \rightarrow RHS) = \frac{Support(LHS \cup RHS)}{Support(LHS)Support(RHS)} \tag{12}$$

As suggested by Bu and Cho (2023), the Bayesian network is constructed to present causal relationships between symbols based on association rules. In addition, this model is useful for avoiding collisions between rules on a power demand condition. We improve it by incorporating domain knowledge when constructing the network for a better

interpretation. In this paper, we find causal relationships that occur in energy demand from literature.

Structure Learning. Bayesian network is defined by equation (13), where N, e, and Pa represent nodes, evidence, and parents, respectively. The posterior probability N given evidence e is calculated by $P(N|e)$. Using Bayesian network, we present a causal effect of variables A to B using directed arc as $A \rightarrow B$.

$$P(N|e) = \prod P(N|Pa(N)) \times e = \prod P(N|Pa(N)) \prod_{e_i \in e} e_i \tag{13}$$

Determining the structure of Bayesian network is inherently challenging due to the complex interdependencies between variables. Different structural configurations can lead to significantly different interpretations and insights, making the choice of structure crucial for accurate model outcomes (Bu and Cho, 2023). To address this issue, we actively incorporate domain knowledge into the learning process. This integration not only guides the structural design but also significantly improves the interpretability of our model, ensuring that the results are both meaningful and actionable in practical scenarios. Fig. 2 depicts the full pipeline of structure learning.

Buys et al. developed a structural model to identify critical factors of peak demand from complex, interacting factors and grouped variables by themes such as house type, physical environment, and appliances to explain the model's behavior patterns (Buys et al., 2015). In contrast, Khan created a model that provides causal relationships between factors influencing energy peak demand in two different views—daily and seasonal—highlighting that peak demand occurs in seasonal patterns (Khan, 2021). Using that knowledge, four themes are extracted from our observed variables (Table 2) from UCI household dataset. The variables are connected from physical environment → appliance usage → electricity consumption → peak demand. Since the DLPs contribute directly to the prediction result based on the observation window, we assume that every DLP has an independent connection to the peak demand.

To construct the Bayesian network using association rules and domain knowledge, we employ a structure learning algorithm using software Genie Academic v4.1. Our method (Fig. 2) differs in that we apply domain knowledge to avoid *unusual relationships* by implementing a *domain knowledge (DK) restriction* based on the theme. One of the restrictions is that the node cannot have a causal relationship direction from child to parent based on the known DK. For instance, rather than forming a relationship from peak warning to global active power, our method intends to establish a relationship from global active power to peak demand. This is achieved by prohibiting reverse connections between themed variables and by only evaluating the strength of the connections in the appropriate manner. Furthermore, since DLPs directly impact the prediction results, it is assumed that there is a direct connection between DLPs and the prediction outcomes in EDPs.

Table 2
Grouped causal themes with the corresponding observed variables.

Theme	Observed Variables
Physical environment	Date, time, season, day of week
Energy demand pattern	Peak warning, no significant change, lower than usual
Electricity consumption	Global reactive power, voltage, global intensity, global active power
Appliance usage (appliance)	Sub-metering 1 (kitchen), sub-metering 2 (laundry), sub-metering 3 (climate control), sub-metering 4 (other)

Fig. 3 shows the Bayesian network designed with the extracted symbols as mentioned previously. The network consists of three main structures, representing three different power conditions in the t step ahead after the observation window, five variable themes, twelve observed variables, and four deep learning parameters. The Bayesian network constructed using the rule from domain knowledge helps it learn interpretable structures. For example, no notable change is affected by physical environment and electricity consumption. While the current structure reflects the characteristics of the dataset used, deploying the model in different building types and regions may require adjusting the network to account for regional differences in climate, infrastructure, and consumption behavior. The Bayesian network should be retrained or adapted with localized data and domain knowledge to maintain the interpretability and reliability of causal explanation.

Deep Learning Parameter Interpretation. Since our method uses DLP to explain EDP, it is required for user to be able to interpret S_{CAM} and S_{ATT}. To interpret the DLPs, clustering analysis (Satre-Meloy et al., 2020) with cosine similarity threshold are utilized to extract patterns based on the corresponding DLP cluster. Furthermore, in the clustering phase of S_{ATT}, instead of using raw data, the cumulative of each feature is used to capture temporal variations in the dataset.

The S_{CAM} obtained after the clustering process contains useful information about specific power state conditions. For example, cluster A displays an increasing pattern in the climate control device, while cluster B shows the opposite. Although this information is naturally grouped according to similar patterns, we recognize that only several features may exhibit consistent patterns. To identify these useful features, we utilize cosine similarity (Manzo-et al., 2013) for each feature within its cluster and then apply a threshold to filter them. These filtered features are then analyzed and used to enhance the interpretability of our model. Given observation data $X_{c,f}$ of feature f at cluster c, the average cosine similarity is given by:

$$AverageCosineSimilarity = Mean\left(cosine\left(x_{cf_i}, x_{cf_j}\right)\right) \tag{14}$$

where $i = \{0,1,..,N-1\}, j = \{0,1,..,N-1\}, i \neq j$.

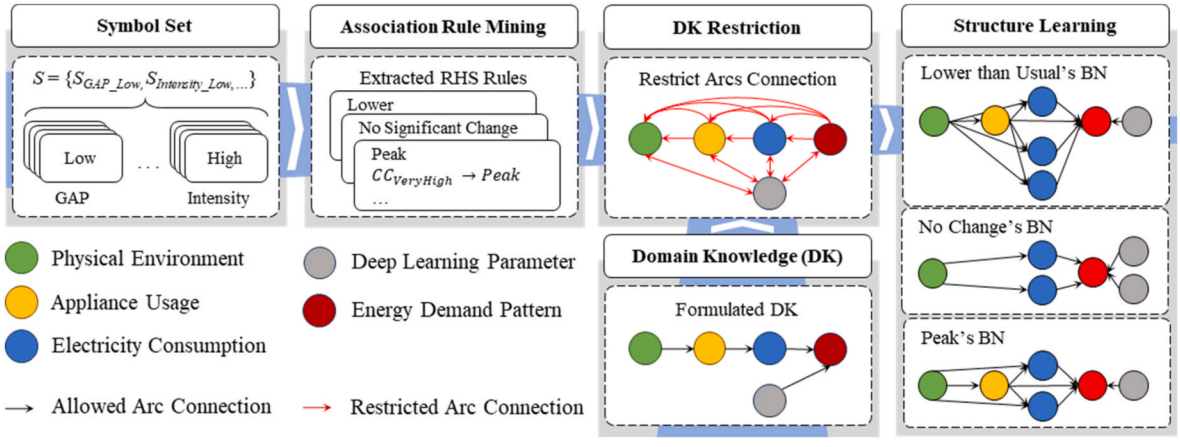


Fig. 2. Structure learning with domain knowledge.

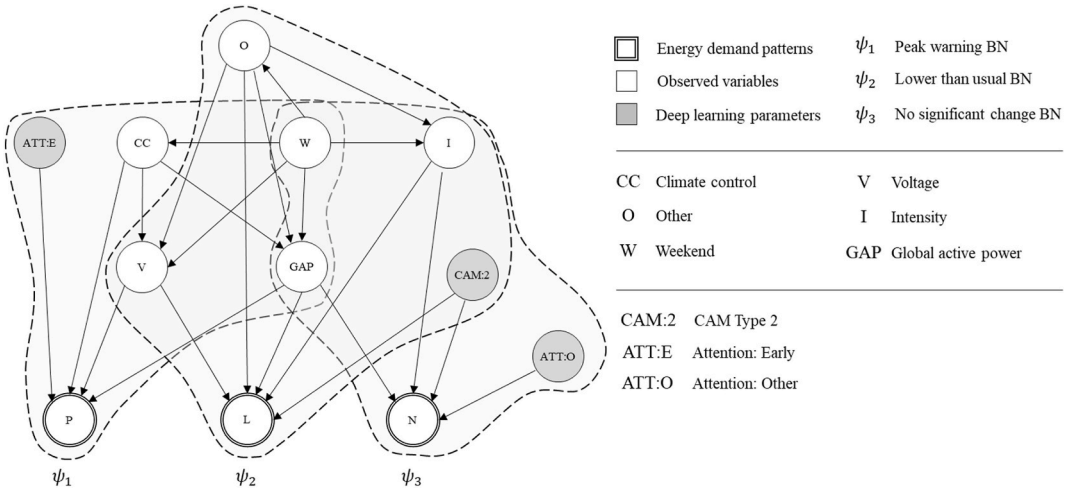


Fig. 3. Bayesian network for three representative energy prediction patterns in step ahead.

Lastly, to get interpretable S_{ATT}, we perform cumulative clustering proposed by Satre-Meloy et al. which is useful to extract temporal variation of the attention vector (Satre-Meloy et al., 2020). To conduct this process, at first, the attention vectors are normalized by subtracting the minimum demand from each hourly value (referred to as 'de-minning'). Then, each hour's consumption is divided by the total 'de-minned' consumption. Given attention vector A_c at cluster c, the cumulative data is given by the following equation (15).

$$C(A_c) = \sum_{i=1}^{N} a_i \tag{15}$$

This is applied to all features of the cluster members. Finally, the preprocessed attention vector is clustered using K-means algorithm and visualized for better interpretability.

To conclude, Fig. 4 summarizes the algorithm of the proposed method, starting with training the prediction model, symbol extraction, Bayesian network initialization and finally returning the Bayesian network as causal explanation framework.

4. Experimental results

4.1. Dataset

We evaluate the proposed method with two real datasets such as the UCI individual household electricity dataset and the REFIT dataset. The former contains 2,075,259 measurements gathered in 1-min time resolution with eight attributes, such as collection time, active power, reactive power, voltage, intensity, and four sub-meterings for kitchen, laundry, climate control, and other appliances. The latter consists of 5,733,526 measurements, sampled with 8-s intervals. This dataset has

Algorithm 1: Energy demand prediction and explanation using Bayesian network

Input : Sequence dataset X

Output : Prediction result $\hat{y}$ and Bayesian network as causal explanation

Step 1. Initialize parallel CNN-LSTM with attention

Train the prediction model using X

Obtain prediction result $\hat{y}$ (eq. 4)

Step 2. Extract symbols (S)

Extract energy demand patterns using f_{EDP} (eq. 7)

Extract different states of energy usage using f_{usage} (eq. 9)

Extract clusters from attention weights a (eq. 1)

Extract clusters from $Activation\ Map$ (eq. 10)

Step 3. Initialize Bayesian network (Structure Learning)

Run association rule mining on S from step 2 resulting in S_{assoc}

Group variables in S based on domain knowledge

Restrict arcs connection with hierarchical rule

Learn Bayesian network structure with domain knowledge EDP using variables in S_{assoc}

Step 4. Produce causal explanation

Calculate the explanation for each f_{EDP} by setting target to true in Bayesian network

Fig. 4. The algorithm of the proposed method.

11 attributes, such as sample time, aggregate power or total load, and nine different household appliances. In addition, feature extraction is performed by extracting day, month, season, and weekend to improve the prediction result with better interpretability.

4.2. Prediction accuracy

First, we compare the performance of the proposed model with other methods. We use mean squared error (MSE), root mean square error (RMSE), and mean absolute error (MAE) to compare the gap between the prediction result and the real value in the time series. The lower the value, the better the model's performance.

$$MSE = \frac{1}{n}\sum_{i=1}^{N}(y_i - \hat{y}_i)^2 \tag{16}$$

$$RMSE = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y}_i)^2} \tag{17}$$

$$MAE = \frac{\sum_{i=1}^{N}|y_i - \hat{y}_i|}{n} \tag{18}$$

We compare standalone CNN and LSTM models with other time series prediction methods and hybrid models, including the proposed method. The experiments are conducted at a 15-min time resolution and evaluated using multiple metrics. As shown in Table 3, the proposed method not only outperforms the standalone models but also the other model categories. This suggests that combining spatial features from CNN and temporal features from LSTM significantly enhances energy prediction accuracy.

Building on these findings, we compare the proposed CNN-LSTM-attention model with state-of-the-art methods. As shown in Table 4, our model consistently achieves superior performance across four different time resolutions while maintaining statistical significance in the *t*-test, outperforming previous models (Kim and Cho, 2019; Bu and Cho, 2020), especially serial configuration with average 34.84 % and 13.63 % improvement in UCI and REFIT household datasets, respectively. Additionally, as shown in Fig. 5(a), the model effectively detects sharp peaks, which is particularly useful for addressing peak demand issues. The improved results (Fig. 5(b) and Table 4) stem from the richer and more detailed spatial-temporal features extracted in parallel rather than in a serial configuration.

4.3. Explainability of prediction results

Consistency is an important aspect in explanation model (Pillai and Pirsiavash, 2021) which accurately reflects the underlying reasoning of the AI systems to build trust and ensure reliability. To assess this, we evaluate the structure of the constructed model (Fig. 3) using cosine similarity on both the training and test sets. This evaluation compares explanation result from Partial Dependence (PD) Variance (Greenwell et al., 2018), LIME, SHAP and our method, as outlined in the previous study (Bu and Cho, 2023). A higher similarity indicates a consistent model under different data conditions, suggesting consistent feature attribution (Lundberg and Lee, 2017). As shown in Table 5, our model consistently achieves higher similarity scores across all energy demand conditions compared to PD Variance, LIME and SHAP. These results further demonstrate that the proposed method is robust against overfitting.

Table 4
Comparison of the proposed method with the other CNN-LSTM models (MSE).

Time Resolution	S-CNNLSTM	S-CNNLSTM$_{Att}$	P-CNNLSTM$_{Att}$ (ours)
UCI Household Dataset			
15M	0.00364	0.00330	**0.00307**
30M	0.00649	0.00636	**0.00431**
1H	0.00797	0.00704	**0.00386**
1D	0.00624	0.00693	**0.00312**
REFIT Household Dataset			
15M	0.01485	0.01455	**0.01433**
30M	0.00326	0.00415	**0.00303**
1H	0.01151	0.01143	**0.00886**
1D	0.01823	0.01852	**0.01787**

Interpretable Deep Learning Parameters. We employ the clustering method on the preprocessed cumulative data and select the best cluster with the elbow method based on the within-cluster sum squares (WCSS). A threshold is applied after measuring the similarity to identify features that have consistent patterns. Fig. 6 shows the clustering and similarity measurement with threshold 0.5, with left cluster 1, and right cluster 2 from activation map. From this result, we can conclude that the CAM type 1 is activated because there are increasing patterns in the laundry, climate control and other unspecified devices. Meanwhile, CAM type 2 is activated because of the decreasing patterns in laundry, kitchen, and climate control usages. This result is very intuitive when used to interpret Bayesian network structure. For example, 'lower than usual' and 'no significant change' are caused by CAM type 2. It indicates that a lower consumption of power would be achieved if the household occupants decreased their appliance usage, such as laundry, kitchen, and climate control.

The second valuable information from deep learning parameters is the attention vector. This vector highlights important parts of the specific time that carries more information (Heidari and Khovalyg, 2020). For example, the attention vector assigns higher weights to the times displaying an increasing pattern, rather than to those with a decreasing pattern. Utilizing the method described in the previous section, three clusters of attention vectors have been identified and visualized in Fig. 7. Late attention highlights an increasing pattern of power usage, while early attention indicates a decreasing trend. Meanwhile, the 'other' category captures undefined patterns, potentially reflecting fluctuations or stability in power demand.

According to the learned Bayesian network, the 'no significant

Table 3
Comparison with other models across three different performance metrics.

Category	Method	UCI Household Dataset			Refit Dataset		
		MSE	RMSE	MAE	MSE	RMSE	MAE
Ensemble model	XGBoost (Sim et al., 2022)	0.01440	0.12000	0.08078	0.01739	0.12937	0.07820
Statistical model	ARIMA	0.01307	0.11306	0.09022	0.01582	0.12349	0.06858
Nonlinear dynamic model	SINDy (Brunton et al., 2016)	0.00596	0.07701	0.04701	0.03107	0.17627	0.09226
Machine learning model	SVR (Sim et al., 2022)	0.01839	0.13559	0.12223	0.02531	0.14981	0.10156
	ANN (Białek et al., 2022)	0.00434	0.06588	0.04437	0.02010	0.14177	0.09599
	CNN	0.01982	0.14077	0.11784	0.04469	0.21141	0.16713
	LSTM (Sim et al., 2022)	0.00373	0.06102	0.03961	0.01447	0.12017	0.06464
Hybrid model	S-CNN-LSTM (Kim and Cho, 2019)	0.00364	0,06033	0.03895	0.01485	0.12185	0.06409
	S-CNNLSTMAtt (Bu and Cho, 2020)	0.00330	0.05744	0.03904	0.01455	0.12062	0.06407
	P-CNNLSTMAtt (Ours)	**0.00307**	**0.05541**	**0.03628**	**0.01433**	**0.11971**	**0.06369**

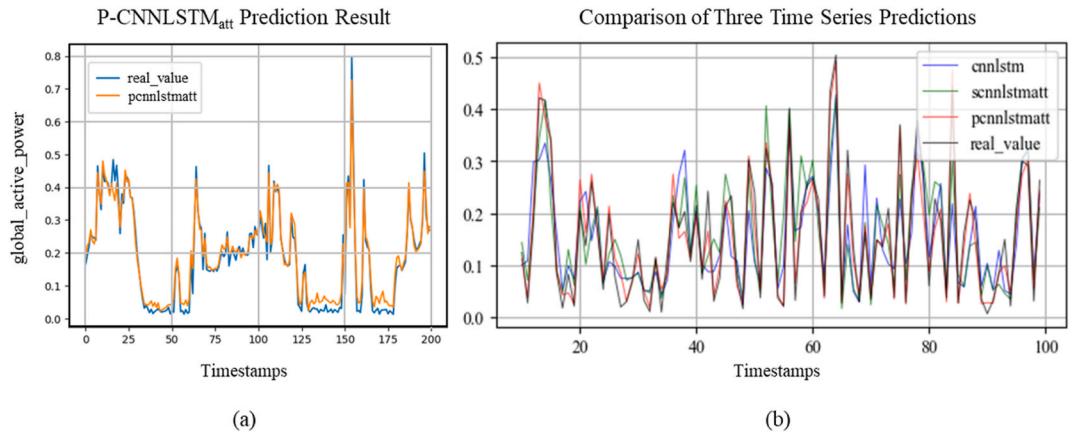


Fig. 5. (a) Prediction performance of parallel CNN-LSTMs with attention and (b) comparison with other CNN-LSTM methods.

Table 5
Consistency comparison of our method and Shapley value to provide global explanation using similarity measurements.

Energy Demand Pattern	PD Variance	LIME	SHAP	Ours
Lower than usual	0.80629	0.69221	0.95310	**0.99974**
No significant change	0.92643	0.75056	0.96091	**0.99983**
Peak warning	0.96062	0.70791	0.95840	**0.99940**

change' and 'peak warning' EDP classifications are influenced by the 'other' and 'early' attention types, respectively. The first EDP state occurs when power levels are stable. Interestingly, our deep learning model captures a decreasing pattern before a peak warning occurs. This indicates that the state of peak warning is likely to happen when overall demand is lower, followed by a sudden jump in energy usage that increases the demand. To conclude, these results show that attention vector is important to understand common patterns that affect some power state based on historical information (see Table 6 for a summary

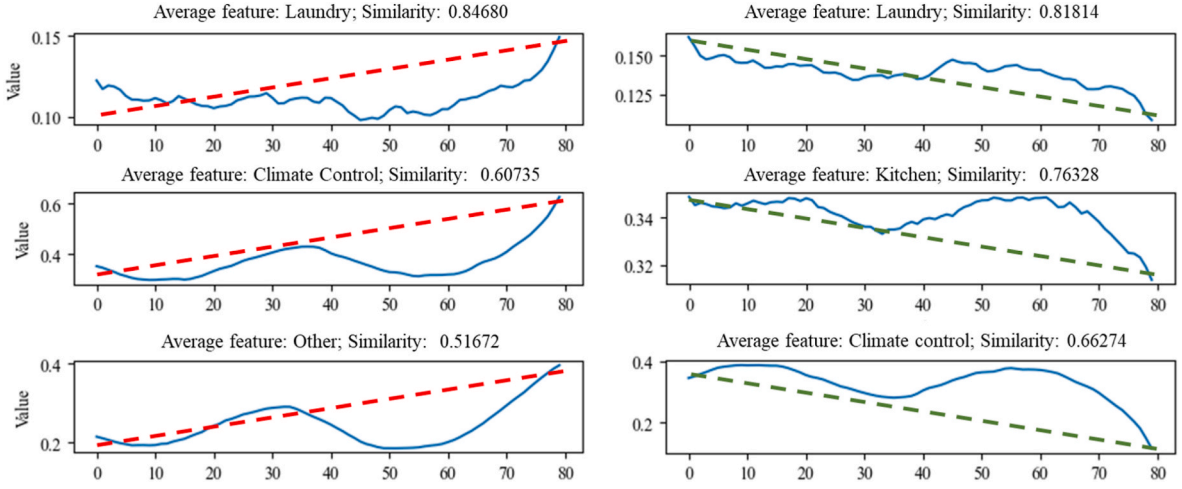


Fig. 6. Filtered patterns of two clusters of activation maps (left: type 1, right: type 2) using average cosine similarity.

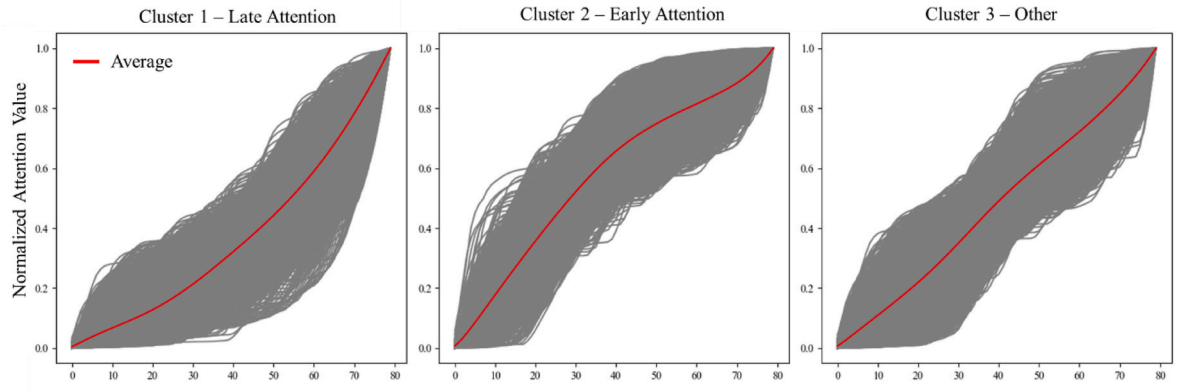


Fig. 7. Three different patterns of attention vectors captured by cumulative clustering technique.

of DLPs interpretations).

Causal Explanations for Three EDPs. We evaluate the comprehensiveness of our method (Rajapaksha and Bergmeir, 2022) for causal explanation by presenting a summary of the Bayesian network and conducting a sensitivity analysis using tornado chart. We can provide relevant explanations and implications for household occupants, offering actionable insight to avoid peak load events, due to the global explanation capabilities of the Bayesian network. This explanation scenario is based on the energy domain, thus reflecting a real-world approach to energy saving. The results expect to improve the domain of consumer behavior management, enhancing both information awareness and demand management (Mosannenzadeh et al., 2017).

Peak warning. In Fig. 8, early attention is expected to hit 53 %, meaning that the global decreasing pattern would trigger a peak. Furthermore, a very high state of global active power and climate control is observed, with probabilities of 48 % and 50 % respectively. It is also shown that there is only a 33 % chance that weekends contribute to the peak. The sensitivity analysis in Fig. 9 reveals that very high usage of climate control has the greatest influence on peak. In contrast, the low state of climate control reduces the probability of peak warning to 34 %.

Actionable insight: Given the high sensitivity of the peak warning state to the usage of climate control devices, household occupants are advised to carefully avoid setting these devices to very high levels, as this could result in excessive power consumption.

Lower than usual. In contrast to peak times, lower state occurs due to an 85 % decrease in activity involving kitchen, laundry, and climate control devices, as indicated by CAM type 2. Our model suggests that reducing the use of these devices can significantly decrease power consumption and lead to energy savings. This reduction is characterized by a low intensity (42 %) and a decrease in global active power consumption (39 %). Additionally, based on sensitivity analysis, our model captures weekday and very high usage of 'other' appliances as the opposite cause of lower EDP.

Actionable insight: Household occupants should avoid the very high usage of 'other' appliances and instead use them at medium intensity to save energy. Additionally, by reducing the usage of kitchen, laundry, and climate control devices, significant energy savings can be achieved.

No significant change. Based on the Bayesian network, a stable state of energy consumption is achieved when the global active power and intensity are maintained at steady levels, as evidenced by the even distribution of *low*, *medium* and *high* states. Additionally, undefined fluctuations or stable conditions are also captured by the 'other' attention category. Furthermore, a decreasing pattern of laundry, kitchen, and climate control contributes to maintaining stable energy consumption, as shown by CAM 2.

Actionable insight: Using households' appliances without a sudden jump in energy consumption will lead to stability. Similar to the lower EDP scenario, avoiding increased usage of kitchen, laundry, and climate control devices also help preserve this stability.

Table 6
Summary of deep learning parameters with their interpretation.

Deep Learning Parameter	Meaning/Interpretation
CAM 1	**Increasing** pattern of **laundry**, **climate control**, and **other equipment** in time observation
CAM 2	**Decreasing** pattern of **laundry**, **kitchen**, and **climate control** equipment in time observation
Late Attention	**Increasing** pattern for **overall energy demand** in time observation
Early Attention	**Decreasing** pattern for **overall energy demand** in time observation
Other Attention	**Stable or fluctuation** pattern for overall energy demand in time observation

4.4. Computation overhead

In this section we provide an additional experiment to compare the potential computation overhead produced by our explainable method compared to SHAP and LIME. To compare the performance, we use 10 different sample sizes from 10 to 100. For the proposed method utilizing the Bayesian Network, we calculate the time required from structure learning to inference, while for LIME and SHAP, we compute the time required to assign feature contributions for each sample size. Table 7 presents the execution time results, and Fig. 10 shows the comparison.

The results indicate that SHAP has the highest computational cost, with execution time increasing exponentially as the sample size grows, reaching 28,035 s (~7 h) for 100 samples. This is due to SHAP's need to generate multiple feature subsets and rerun the prediction model for each, significantly increasing complexity (Lundberg et al., 2020). LIME, while more efficient than SHAP, still requires more time than BN, as it perturbs samples and fits a surrogate model for each prediction. In contrast, BN remains computationally efficient, making it a scalable alternative for explainability.

Table 8 shows the execution time of structure learning and inference time of the Bayesian network. Structure learning is performed once on the dataset and takes considerably more time (~433.61 s) compared to inference, which averages 1.3 s per data instance. Moreover, the structure learning is expected to increase due to the number of data and variables (Huang and Suzuki, 2024). However, since structure learning is conducted only once during model initialization, the low inference time makes the approach efficient and well suited for repeated real-time queries.

In addition, we also evaluate running time on three different configurations of CNN-LSTM variants, presented in section 4.2. For this evaluation, each model is required to predict $(t+1)$ condition after observing 80 data points from previous observation, with 1 h time resolution. As shown in Table 9, more complex models tend to have a longer running time. Combining these results with those from Table 4, an accurate model comes with the tradeoff of a higher inference time. However, when considering inference for the next hour, the inference time is still negligible, where it only takes approximately ¼ seconds.

4.5. Ablation study

We conduct an ablation study to evaluate the contribution of each architectural component (Fig. 11) in our CNN-LSTM with attention model. Table 10 shows the model components. The full model (p-c-l-a) have achieved the best performance across all metrics, demonstrating the effectiveness of combining CNN, LSTM, and attention in parallel configuration. Removing attention mechanism (p-c-l) leads to a slight drop in performance, which indicates its marginal contribution. Serializing the CNN and LSTM (s-c-l-a) reduces the model's performance, implying that parallel model captures better patterns and reduces the abstraction in the serial model.

We conduct additional ablation studies on the explanation module by removing deep learning parameters (DLPs) from the Bayesian network. As presented in Fig. 12, model with DLP achieves higher similarity score, indicating more stable and consistent explanations. It demonstrates the positive impact of incorporating DLP into the explanation.

Furthermore, we investigate the performance of the Bayesian network under different discrete states after quantization (q) using normalized accuracy. As shown in Fig. 13, the performance of the Bayesian network improves slightly as the number of discrete states increases. However, increasing the number of states may harm interpretability, and explanations become harder to interpret. Therefore, keeping the number of discrete states moderately can provide a proper balance between accuracy and interpretability.

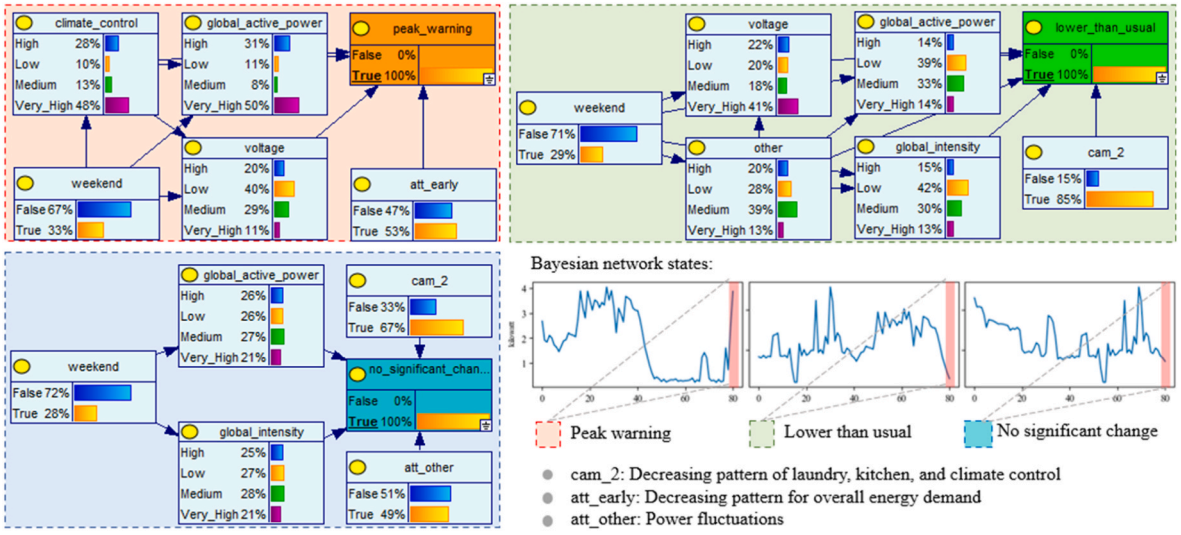


Fig. 8. Bayesian network with activated state for each EDP.

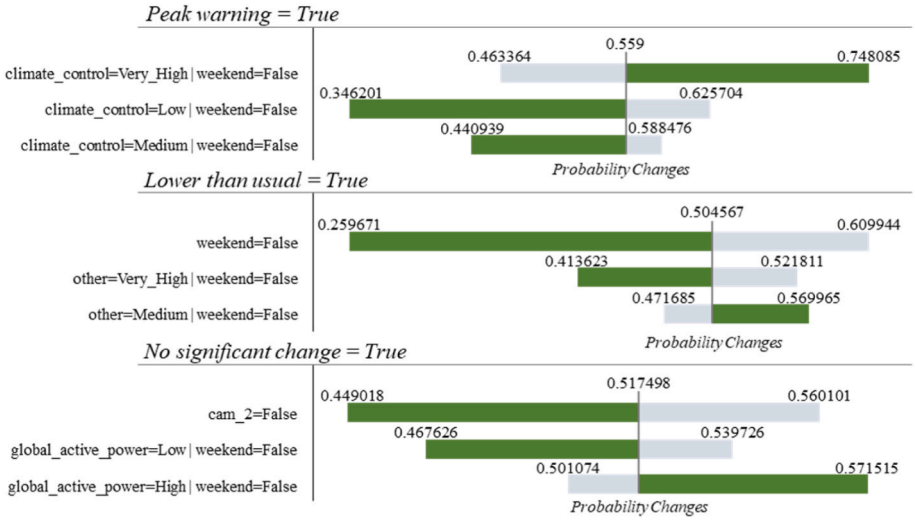


Fig. 9. Sensitivity analysis for the top three variables across three different power states, with the activated state set to True. Green bars represent the probability of the power demand state after adjusting the specified variables (shown at the left side of the chart).

Table 7
Detailed execution time of ours and SHAP in seconds.

Number of samples	Peak warning			Lower than usual			No significant change		
	Ours	LIME	SHAP	Ours	LIME	SHAP	Ours	LIME	SHAP
10	1.33	423.18	276.10	34.84	391.60	280.24	7.96	380.81	279.86
20	1.03	825.68	1111.88	75.89	784.09	1139.26	84.19	747.14	1112.59
30	1.12	1278.01	2469.99	83.19	1194.38	2518.18	87.32	1163.26	2504.12
40	4.17	1689.97	4405.23	74.59	1586.76	4579.30	77.63	1536.23	4483.85
50	35.41	2043.63	6865.26	92.41	1905.70	7123.45	78.34	1903.77	6993.48
60	34.92	2499.03	9921.59	92.32	2287.22	10129.52	80.32	2266.80	9860.84
70	38.08	2934.85	13423.01	76.07	2662.96	13774.26	81.12	2583.06	13475.34
80	38.54	3346.10	17380.71	77.21	3111.94	17788.01	75.45	2955.13	17584.21
90	41.38	3703.97	22027.83	73.57	3493.91	22465.37	77.12	3294.90	22101.64
100	35.50	4033.92	27310.59	79.78	3868.97	28035.31	77.97	3783.67	27456.40

4.6. Model limitation

In this section, we explore the limitations of the proposed model. Fig. 14 shows its performance compared to other CNN-LSTM configurations with feature perturbations. We introduce noise at various levels in the input values and measure the relative increase of prediction error. Higher error indicates the model is prone to noise. As shown in the figure, serial models (s-cnn-lstm-att and s-cnn-lstm) are more robust to the noise. Furthermore, adding attention can improve the model performance (p-cnn-lstm-att and s-cnn-lstm-att), implying that this

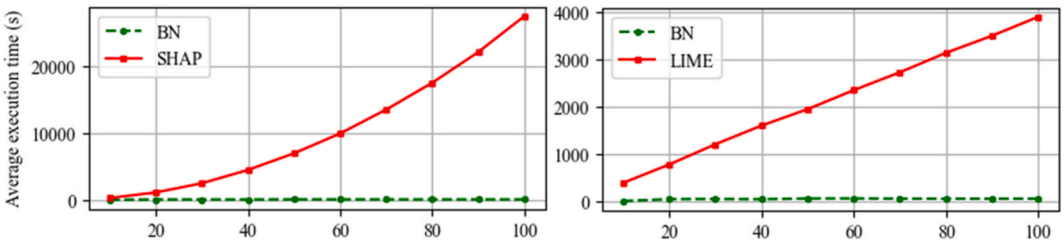


Fig. 10. Execution time comparison of SHAP (left), LIME (right), and our method across different sample sizes.

Table 8
Runtime comparison of structure learning and inference (seconds).

Step	Time
Structure learning	433.61
Inference	1.30

Table 9
Execution time comparison on different CNN-LSTM model configurations in milliseconds (ms) with average and standard deviation.

Metrics	S-CNNLSTM	S-CNNLSTM$_{Att}$	P-CNNLSTM$_{Att}$ (ours)
Average (ms)	186.065	236.639	260.857
Stdev (ms)	2.773	14.878	9.831

component plays a key role in mitigating the effect of noisy input.

5. Discussion

This study was motivated by the need for accurate and interpretable energy demand forecasting models that can support or influence energy-saving behavior among household occupants. Deep learning models such as CNN-LSTM have proven effective in capturing temporal and spatial features but lack interpretability especially in high stake domains. To solve it, we have proposed a method that integrates a parallel CNN-LSTM-attention with causal explanation which captures relationship among input features, deep learning parameters and prediction result. As a result, the proposed method not only provides accurate prediction, but also demonstrates the ability to generate interpretable causal explanation that can support both user understanding and informed decision-making.

The evaluation has shown that the proposed model consistently outperformed other time series prediction models, including baseline CNN-LSTM-attention in serial configuration, with up to 34 % improvement in prediction accuracy on the household dataset. Moreover, ablation study uncovers the parallel structure with attention contributes to a better performance than the baseline. Furthermore, the integration of CAM and ATT enables the model to provide interpretable patterns aligned with domain expectation. For example, reduction of appliance usage as suggested by CAM type 2 could increase the probability of lower energy consumption. These results highlight the effectiveness of combining architectural innovations with a causal explanation framework to improve both performance and interpretability.

Despite those findings, we acknowledge that this work has not been explored for $(t+n)$ prediction setting, which indeed could introduce both modeling and interpretability challenges. From a prediction standpoint, longer-term forecasts require the model to effectively capture seasonal patterns, longer dependencies, and effective temporal structure. Moreover, as the horizon extends, the feature space becomes less certain and not directly observable, which increases the complexity of the task. Meanwhile, in terms of causal explanation, the reliability of

Table 10
Description of model components.

Component	Description
p	Parallel configuration of CNN and LSTM
s	Serial configuration of CNN and LSTM
c	CNN module
l	LSTM module
a	Attention module

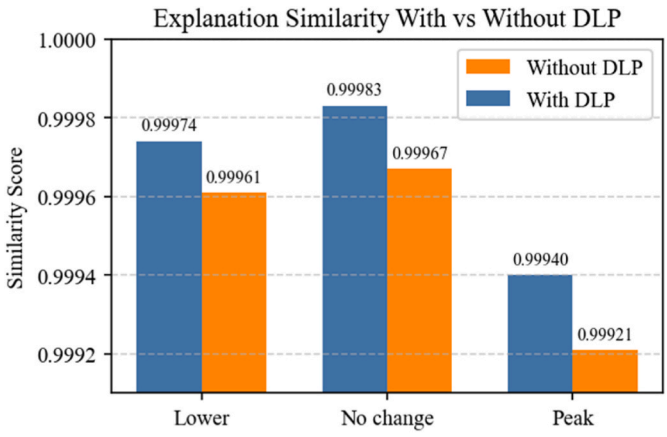


Fig. 12. Ablation study results on DLP integration in Bayesian network.

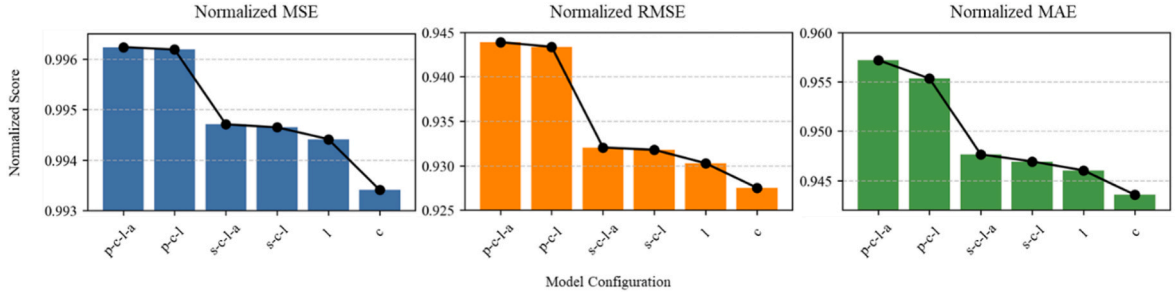


Fig. 11. Ablation study results on different model configurations.

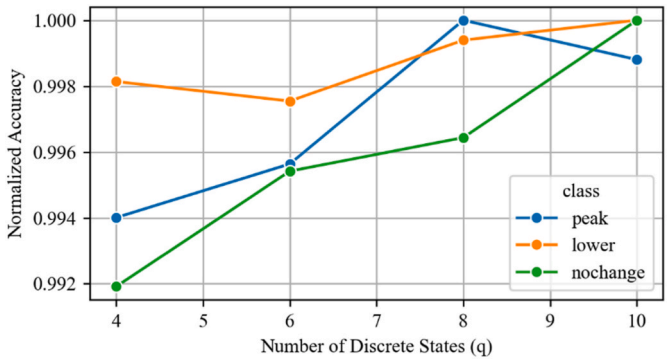


Fig. 13. Normalized model performance under different discrete states.

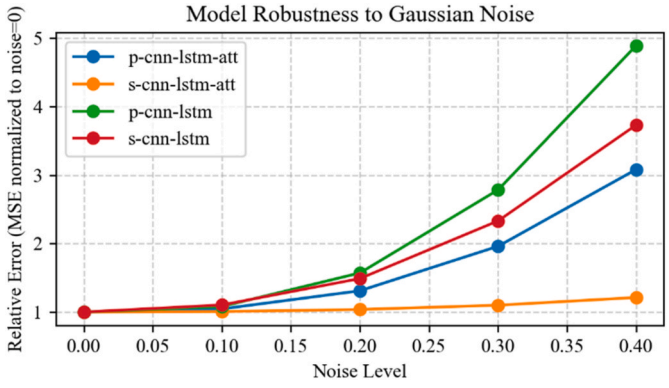


Fig. 14. Experiment results of model robustness to noise.

explanations depends on how well the model captures meaningful and stable features over a longer temporal context. For example, understanding the usage of an entire upcoming month may require seasonal knowledge. Therefore, the model must identify features that are robust across time, not just locally relevant at t.

Future work can focus on these directions: First, extending the proposed method beyond one step ahead $(t+1)$ forecasting to support multi-step $(t+n)$, which requires improved handling of long-term dependencies and stable generation of explanation over time. Second, it is worth exploring the capabilities of the proposed method in the different domains beyond household energy consumption. Lastly, the explanation module could be used in other deep learning architecture, for example Transformer model.

6. Concluding remarks

In this paper, we have proposed parallel CNN-LSTM with attention to give a prediction in $(t+1)$ step ahead, which outperforms the statistical, machine learning, ensemble, and hybrid CNN-LSTM methods. Furthermore, a method to explain three energy demand prediction states using Bayesian network is provided. We improve the interpretation of the Bayesian network so it can be easier to understand for household occupants by incorporating domain knowledge and deep learning parameter analysis, aligned with interpretability criteria (Mavrepis et al., 2024). It is found that deep learning can capture spatial and temporal data through parallel configuration of CNN-LSTM with attention. This separation not only increases the prediction model's performance but also improves its interpretability through further analysis.

Furthermore, we have demonstrated that our method is able to provide explanations in the energy domain, especially in consumer behavior management (Mosannenzadeh et al., 2017). This is achievable because Bayesian networks offer various analytical tools in addition to their structure to provide causal explanations. The proposed method presents an innovative approach that bridges accurate forecasting and interpretability challenge and provides actionable insight in energy system, which is an important goal in AI for sustainability.

In the future, we plan to explore additional capabilities of the Bayesian network, such as automated structure learning, dynamic structure to handle new data, and extended causal explanation features. In addition, we will improve our method so that it is not limited to a specific domain but can have a broader impact and generalization. Especially, in the biomedical domain, it can enhance decision-making by providing transparent insights into patient conditions, and in industrial domain, it can monitor and explain parameter changes in complex system, such as gas pipelines (Zapukhliak et al., 2019).

CRediT authorship contribution statement

Gatum Erlangga: Writing – original draft, Visualization, Methodology, Formal analysis, Data curation. **Sung-Bae Cho:** Writing – review & editing, Validation, Supervision, Resources, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Sung-Bae Cho reports financial support was provided by Institute of Information & Communications Technology Planning & Evaluation and Air Force Office of Scientific Research. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This work was supported by the Yonsei Fellow Program funded by Lee Youn Jae, IITP grant funded by the Korea government (MSIT) (No. RS-2022-II220113, Developing a Sustainable Collaborative Multi-modal Lifelong Learning Framework), and Air Force Defense Research Sciences Program funded by Air Force Office of Scientific Research.

Data availability

Data will be made available on request.

References

Afzalan, M., Jazizadeh, F., Eldardiry, H., 2021. Two-stage clustering of household electricity load shapes for improved temporal pattern representation. IEEE Access 9, 151667–151680.
Alduailij, M.A., Petri, I., Rana, O., Alduailij, M.A., Aldawood, A.S., 2021. Forecasting peak energy demand for smart buildings. J. Supercomput. 77, 6356–6380.
Amiri, S.S., Mottahedi, S., Lee, E.R., Hoque, S., 2021. Peeking inside the black-box: explainable machine learning applied to household transportation energy consumption, Computers. Environment and Urban Systems 88, 101647.
Bertoldi, P., 2022. Policies for energy conservation and sufficiency: review of existing policies and recommendations for new and effective policies in OECD countries. Energy Build. 264, 112075.
Białek, J., Bujalski, W., Wojdan, K., Guzek, M., Kurek, T., 2022. Dataset level explanation of heat demand forecasting ANN with SHAP. Energy 261, 125075.
Brunton, S.L., Proctor, J.L., Kutz, J.N., 2016. Discovering Governing Equations from Data by Sparse Identification of Nonlinear Dynamical Systems, 113. Proceedings of the national academy of sciences, pp. 3932–3937.
Bu, S.-J., Cho, S.-B., 2020. Time series forecasting with multi-headed attention-based deep learning for residential energy consumption. Energies 13, 4722.
Bu, S.-J., Cho, S.-B., 2023. A causally explainable deep learning model with modular Bayesian network for predicting electric energy demand. In: 18th Int. Conf. on Hybrid Artificial Intelligence Systems, pp. 519–532.
Buys, L., Vine, D., Ledwich, G., Bell, J., Mengersen, K., Morris, P., Lewis, J., 2015. A framework for understanding and generating integrated solutions for residential peak energy demand. PLoS One 10, e0121195.
Cakiroglu, C., Demir, S., Ozdemir, M.H., Aylak, B.L., Sariisik, G., Abualigah, L., 2024. Data-driven interpretable ensemble learning methods for the prediction of wind turbine power incorporating SHAP analysis. Expert Syst. Appl. 237, 121464.

Cen, S., Yoo, J.H., Lim, C.G., 2022. Electricity pattern analysis by clustering domestic load profiles using discrete wavelet transform. Energies 15, 1350.
Chen, Y.-C., Wheeler, T.A., Kochenderfer, M.J., 2017. Learning discrete Bayesian networks from continuous data. J. Artif. Intell. Res. 59, 103–132.
Chung, J., Jang, B., 2022. Accurate prediction of electricity consumption using a hybrid CNN-LSTM model based on multivariable data. PLoS One 17, e0278071.
Cunningham, H., Ewart, A., Riggs, L., Huben, R., Sharkey, L., 2023. Sparse autoencoders find highly interpretable features in language models. arXiv preprint arXiv: 2309.08600.
Greenwell, B.M., Boehmke, B.C., McCarthy, A.J., 2018. A simple and effective model-based variable importance measure. iv Preprint arXiv:1805.04755.
Hasan, M., Mifta, Z., Salsabil, N.A., Papiya, S.J., Hossain, M., Roy, P., Farrok, O., 2023. A critical review on control mechanisms, supporting measures, and monitoring systems of microgrids considering large scale integration of renewable energy sources. Energy Rep. 10, 4582–4603.
Heidari, A., Khovalyg, D., 2020. Short-term energy use prediction of solar-assisted water heating system: application case of combined attention-based LSTM and time-series decomposition. Sol. Energy 207, 626–639.
Heo, J., Song, K., Han, S., Lee, D.-E., 2021. Multi-channel convolutional neural network for integration of meteorological and geographical features in solar power forecasting. Appl. Energy 295, 117083.
Huang, H., Suzuki, J., 2024. An efficient procedure for computing bayesian network structure learning. arXiv preprint arXiv:2407.17072.
Ijaz, K., Hussain, Z., Ahmad, J., Ali, S.F., Adnan, M., Khosa, I., 2022. A novel temporal feature selection based LSTM model for electrical short-term load forecasting. IEEE Access 10, 82596–82613.
Khan, I., 2021. Household factors and electrical peak demand: a review for further assessment. Adv. Build. Energy Res. 15, 409–441.
Khan, A.M., Osińska, M., 2023. Comparing forecasting accuracy of selected grey and time series models based on energy consumption in Brazil and India. Expert Syst. Appl. 212, 118840.
Kim, T.-Y., Cho, S.-B., 2019. Predicting residential energy consumption using CNN-LSTM neural networks. Energy 182, 72–81.
Kim, J.-Y., Cho, S.-B., 2023. Predicting residential energy consumption by explainable deep learning with long-term and short-term latent variables. Cybern. Syst. 54, 270–285.
Lu, Y., Zheng, Q., Quinn, D., 2023. Introducing causal inference using Bayesian networks and do-Calculus. Journal of Statistics and Data Science Education 31, 3–17.
Lundberg, S.M., Lee, S.-I., 2017. A unified approach to interpreting model predictions. Adv. Neural Inf. Process. Syst. 30.
Lundberg, S.M., Erion, G., Chen, H., Degrave, A., Prutkin, J.M., Nair, B., Katz, R., Himmelfarb, J., Bansal, N., Lee, S.-I., 2020. From local explanations to global understanding with explainable AI for trees. Nat. Mach. Intell. 2, 56–67.
Manzo-Martínez, A., Camarena-Ibarrola, J.A., 2013. A new and efficient alignment technique by cosine distance. Int. Journal of Combinatorial Optimization Problems and Informatics 4, 12–24.
Martinez Lagunas, A.J., Askarihosni, M., Alimohammadi, N., Dezyanian, A., Nik-Bakht, M., 2021. Discovery of energy performance patterns for residential buildings through machine learning. In: Canadian Society of Civil Engineering Annual Conference, pp. 1–15.
Mavrepis, P., Makridis, G., Fatouros, G., Koukos, V., Separdani, M.M., Kyriazis, D., 2024. XAI for all: can large language models simplify explainable AI? arXiv preprint arXiv: 2401.13110.
Mosannenzadeh, F., Bisello, A., Vaccaro, R., D'Alonzo, V., Hunter, G.W., Vettorato, D., 2017. Smart energy city development: a story told by urban planners. Cities 64, 54–65.
Nagahisarchoghaei, M., Nur, N., Cummins, L., Nur, N., Karimi, M.M., Nandanwar, S., Bhattacharyya, S., Rahimi, S., 2023. An empirical survey on explainable ai technologies: recent trends, use-cases, and categories from technical and application perspectives. Electronics 12, 1092.
Nogueira, A.R., Pugnana, A., Ruggieri, S., Pedreschi, D., Gama, J., 2022. Methods and tools for causal discovery and causal inference. Wiley interdisciplinary reviews: Data Min. Knowl. Discov. 12, e1449.
Park, E., Kim, B., Park, S., Kim, D., 2018. Analysis of the effects of the home energy management system from an open innovation perspective. J Open Innov. Technol. Mark. Complex. 4, 31.
Pillai, V., Pirsiavash, H., 2021. Explainable models with consistent interpretations. In: 35th AAAI Conf. on Artificial Intelligence, pp. 2431–2439.
Pinto, T., Praça, I., Vale, Z., Silva, J., 2021. Ensemble learning for electricity consumption forecasting in office buildings. Neurocomputing 423, 747–755.
Rajapaksha, D., Bergmeir, C., 2022. Limref: local interpretable model agnostic rule-based explanations for forecasting, with an application to electricity smart meter data. In: 36th AAAI Conf. on Artificial Intelligence, pp. 12098–12107.
Ribeiro, M.T., Singh, S., Guestrin, C., 2016a. Model-agnostic interpretability of machine learning. arXiv preprint arXiv:1606.05386.
Ribeiro, M.T., Singh, S., Guestrin, C., 2016b. Why should i trust you?" explaining the predictions of any classifier. In: Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining, pp. 1135–1144.
Richman, J.T., Roberts, R.J., 2023. Assessing spurious correlations in big search data. Forecasting 5, 285–296.
Satre-Meloy, A., Diakonova, M., Grünewald, P., 2020. Cluster analysis and prediction of residential peak demand profiles using occupant activity data. Appl. Energy 260, 114246.
Shi, R., Jiao, Z., 2023. Individual household demand response potential evaluation and identification based on machine learning algorithms. Energy 266, 126505.
Sim, T., Choi, S., Kim, Y., Youn, S.H., Jang, D.-J., Lee, S., Chun, C.-J., 2022. Explainable AI (XAI)-based input variable selection methodology for forecasting energy consumption. Electronics 11, 2947.
Song, S.-Y., Leng, H., 2020. Modeling the household electricity usage behavior and energy-saving management in severely cold regions. Energies 13, 5581.
Stern, P.C., Janda, K.B., Brown, M.A., Steg, L., Vine, E.L., Lutzenhiser, L., 2016. Opportunities and insights for reducing fossil fuel consumption by households and organizations. Nat. Energy 1, 16043.
Tang, X., Chen, A., He, J., 2022. A modelling approach based on Bayesian networks for dam risk analysis: integration of machine learning algorithm and domain knowledge. Int. J. Disaster Risk Reduct. 71, 102818.
Theissler, A., Spinnato, F., Schlegel, U., Guidotti, R., 2022. Explainable AI for time series classification: a review, taxonomy and research directions. IEEE Access 10, 100700–100724.
Tjoa, E., Khok, H.J., Chouhan, T., Guan, C., 2023. Enhancing the confidence of deep learning classifiers via interpretable saliency maps. Neurocomputing 562, 126825.
Wenlong, Z., Tien, N.H., Sibghatullah, A., Asih, D., Soelton, M., Ramli, Y., 2023. Impact of energy efficiency, technology innovation, institutional quality, and trade openness on greenhouse gas emissions in ten Asian economies. Environ. Sci. Pollut. Control Ser. 30, 43024–43039.
Yao, G., Chen, Y., Han, C., Duan, Z., 2024. Research on the decision-making method for the passive design parameters of zero energy houses in severe cold regions based on decision trees. Energies 17, 506.
Yaro, A.S., Maly, F., Prazak, P., Maly, K., 2024. Outlier detection performance of a modified z-score method in time-series RSS observation with hybrid scale estimators. IEEE Access 12, 12785–12796.
Zapukhliak, V., Poberezhny, L., Maruschak, P., Grudz Jr., V., Stasiuk, R., Brezinová, J., Guzanová, A., 2019. Mathematical modeling of unsteady gas transmission system operating conditions under insufficient loading. Energies 12, 1325.
Zhang, C., 2023. Causal Analysis for Generalized Interference Problems. University of California, Los Angeles.
Zhang, Y., Tiňo, P., Leonardis, A., Tang, K., 2021. A survey on neural network interpretability. IEEE Trans. on Emerging Topics in Computational Intelligence 5, 726–742.
Zhou, B., Khosla, A., Lapedriza, A., Oliva, A., Torralba, A., 2016. Learning deep features for discriminative localization. In: IEEE Conf. on Computer Vision and Pattern Recognition, pp. 2921–2929.
Zhuang, D., Gan, V.J., Tekler, Z.D., Chong, A., Tian, S., Shi, X., 2023. Data-driven predictive control for smart HVAC system in IoT-integrated buildings with time-series forecasting and reinforcement learning. Appl. Energy 338, 120936.